Zero-Shot Sensing Radius Generalization in Cooperative Multi-Agent Reinforcement Learning: A Study of InforMARL Across Team Sizes

Colton York

NSF AI-EDGE REU, The Ohio State University

June 24, 2026

1. Motivation

Multi-agent reinforcement learning (MARL) has emerged as a powerful framework for cooperative tasks requiring coordination among teams of autonomous agents [1]. A central challenge in deploying learned MARL policies to real-world settings is generalization – the ability of a trained policy to perform effectively under conditions that differ from those seen during training. One important axis of generalization is sensing radius: the range within which an agent can observe neighbors and obstacles within the environment. In physical systems, sensing range is determined by hardware constraints and environmental conditions and may vary significantly between training and deployment [2].

InforMARL [1] is a graph-based MARL architecture that directly encodes sensing radius into its structure, connecting each agent only to entities within a defined radius ρ via a local graph neural network. While the original work demonstrates strong generalization across varying agent counts, it treats sensing radius as a fixed hyperparameter and does not evaluate whether policies trained at one radius transfer to a different radius at test time. This proposal investigates exactly that question: *does a coordination policy trained under ρ_{train} generalize when deployed under $\rho_{test} \neq \rho_{train}$, and how does this interact with team size?*

2. Background

Multi-agent reinforcement learning addresses settings in which multiple agents must learn cooperative or competitive behaviors through interaction within a shared environment. In cooperative MARL, agents share a joint reward signal and must learn to coordinate their actions to maximize collective performance. A widely adopted training paradigm is centralized training with decentralized execution (CTDE), in which a centralized critic with access to global state information is used during training, while each agent executes its policy using only local observations at test time [3].

InforMARL [1] builds on this paradigm by representing the local neighborhood of each agent as a graph, where nodes correspond to entities – other agents, obstacles, and goals – within a sensing radius ρ, and edges encode pairwise distances. A graph neural network (GNN) with a multi-head attention mechanism aggregates neighborhood information into a fixed-sized vector x_{agg}, which is concatenated with the agent's private observation and passed to the actor network to produce an action. This architecture enables transferability across varying numbers of agents, since the GNN operates on a graph structure rather than fixed-sized state concatenations. However, the sensing radius ρ itself is fixed at training time, and its effect on zero-shot generalization has not been studied.

3. Research Question

This study investigates whether coordination policies trained by InforMARL under a fixed sensing radius ρ_{train} generalize in a zero-shot manner to deployment under a shifted sensing radius $\rho_{test} \neq \rho_{train}$. Specifically, we ask how generalization performance varies as a function of the magnitude and direction of the radius shift, and how team size modulates this effect.

4. Experimental Design

This study employs a factorial experimental design to evaluate zero-shot sensing radius generalization in InforMARL. All experiments use the Navigation task from the original InforMARL environment [1], a 2D continuous world of size 2x2 in which N agents must each navigate to a designated goal while avoiding collisions with other agents and obstacles. Agents operate under partial observability, receiving only entities within their sensing radius ρ as graph inputs. The environment and resulting local graph structure for a focal agent are illustrated in Figure 1.

Training: We train InforMARL policies across 9 configurations formed by crossing 3 team sizes (N = 6, 9, 12) with 3 sensing radii (ρ = 0.3, 0.7, 1.2). These radius values were selected based on a connectivity analysis of the navigation environment prior to any training. For each candidate radius, we simulated 2000 random agent initializations and computed the average number of neighbors visible to each agent As shown in Figure 2, the three selected radii correspond to qualitatively distinct connectivity regimes – sparse (avg. 0.3-0.7 neighbors), moderate (avg 1.4-3.1 neighbors), and dense (avg. 3.1-6.9 neighbors) – across all teams sizes, ensuring that each training condition produces meaningfully different graph structures. Each configuration is trained using RMAPPO [3] as the base algorithm, following the training setup of the original InforMARL paper.

Testing. Each of the 9 trained policies is evaluated under all 3 sensing radii, yielding 27 total evaluation conditions as shown in Table 1. The 9 diagonal conditions (ρ_test = ρ_train) serve as in-distribution baselines, while the 18 off-diagonal conditions constitute the zero-shot generalization tests. No retraining or fine-tuning is performed between training and evaluation – policies are deployed directly under the shifted radius. This design allows us to measure both the magnitude of generalization degradation as a function of radius shift, and its asymmetry – that is, whether policies trained under a small radius generalize better to larger radii than vice versa, and how this pattern interacts with team size.

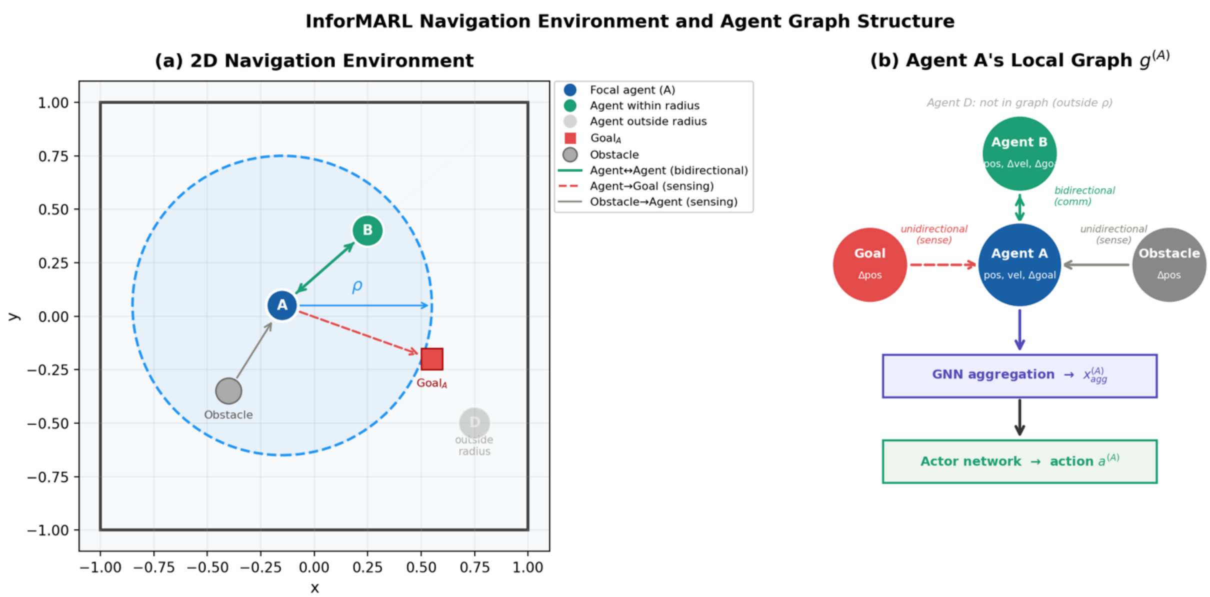

Figure 1 – environment and graph structure diagram

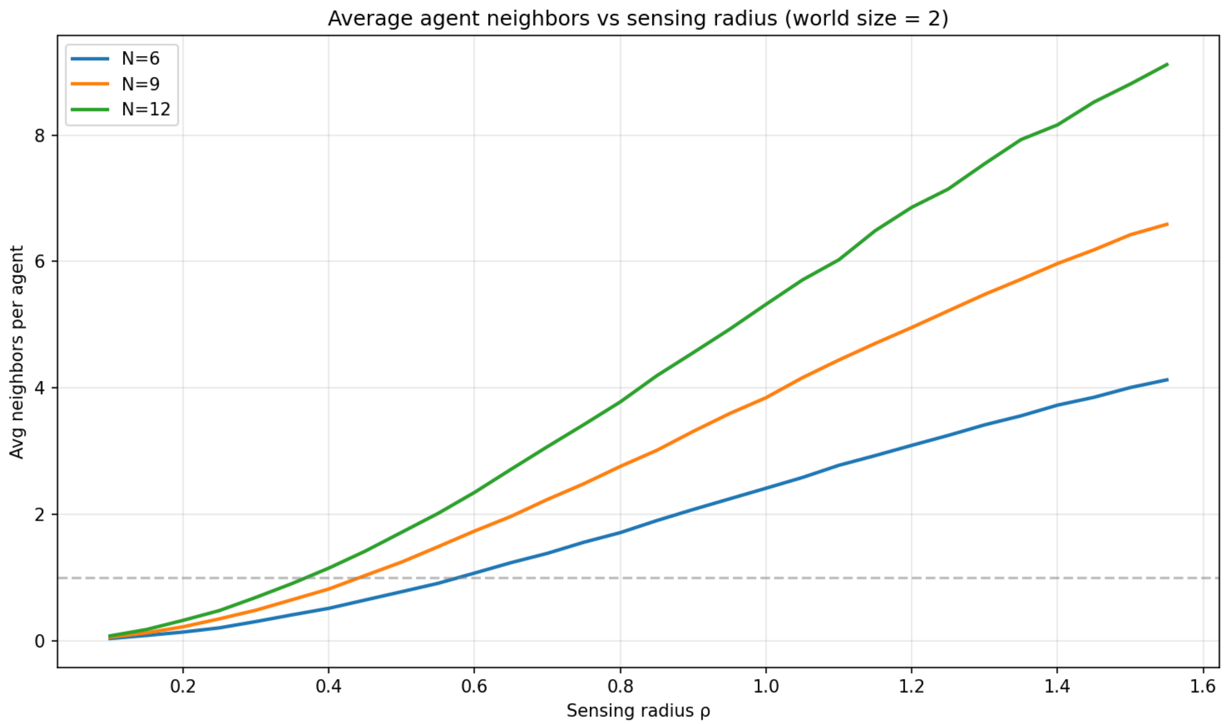

Figure 2 – Connectivity analysis plot

	Test ρ=0.3	Test ρ=0.7	Test ρ=1.2
N=6, Train ρ=0.3	**Baseline**	G	G
N=6, Train ρ=0.7	G	**Baseline**	G
N=6, Train ρ=1.2	G	G	**Baseline**
N=9, Train ρ=0.3	**Baseline**	G	G
N=9, Train ρ=0.7	G	**Baseline**	G
N=9, Train ρ=1.2	G	G	**Baseline**
N=12, Train ρ=0.3	**Baseline**	G	G
N=12, Train ρ=0.7	G	**Baseline**	G
N=12, Train ρ=1.2	G	G	**Baseline**

Table 1 – Full evaluation matrix for zero-shot radius generalization. Rows indicate training configurations, columns indicate test sensing radius. **Baseline** = in-distribution baseline (ρ_train = ρ_test), G = zero-shot generalization condition (ρ_train ≠ ρ_test). 27 total evaluation conditions across 9 training configurations.

5. Metrics

To evaluate zero-shot generalization performance across all 27 evaluation conditions, we report three metrics averaged over multiple evaluation episodes:

Mean agent reward is the primary metric, measured as the cumulative team reward per episode divided by the number of agents N. Each agent receives a per-timestep penalty proportional to its distance to its goal and an additional penalty for collisions with other agents or obstacles. Normalizing by agent count allows direct comparison of performance across team size $N \in \{6, 9, 12\}$, which would otherwise produce incomparable raw reward scales.

Collision rate measures the average number of agent-agent and agent-obstacle collisions per episode. This metric isolates the coordination component of performance – a rise in collision rate under a shifted radius indicated that the learned communication structure has broken down, independent of whether agents are still making progress toward their goal.

Goal completion rate measures the fraction of agents that successfully reach their designated goal within the episode length. This metric isolates the navigation component of performance – a drop in goal completion rate indicates agents are failing to reach their targets, independent of how many collisions occurred,.

Together these three metrics allow us to not only detect generalization failure through team reward, but diagnose its cause – whether shifted radius primarily disrupts coordination, navigation, or both. Each metric is reported as a mean with standard deviation across evaluation in a separate 9x3 results table.

6. Expected Outcomes and Hypotheses

We state three primary hypotheses motivating this study:

H1 – Generalization degrades with radius shift magnitude. We expect mean agent reward to decrease monotonically as the gap between ρ_train and ρ_test increases. Policies trained at $\rho=0.7$ should generalize better to $\rho=0.3$ or $\rho=1.2$ than policies trained at $\rho=0.3$ generalizing to $\rho=1.2$, since the former represents a smaller distributional shift in graph structure.

H2 – Generalization is asymmetric with respect to shift direction. We hypothesize that policies trained under a smaller radius will generalize better to larger radii than vice verse. Agents trained under sparse graphs must learn robust individual navigation with minimal coordination, a skill that remains useful when more neighbors become available. Conversely, agents trained under dense graphs may develop coordination strategies that are over-reliant on rich neighborhood information and collapse when that information is suddenly restricted.

H3 – Generalization degradation increases with team size. As N increases, the coordination problem becomes more complex and agents become more dependent on communication to avoid deadlocks. We expect that larger teams will show steeper performance drops under radius shift, since the consequences of disrupted communication are more severe when more agents are competing for the same space.

These hypotheses are falsifiable and directly testable from the 27-condition evaluation matrix, making this study well-suited as a structured empirical contribution to the MARL generalization literature.

7. Compute Resources

Training 9 independent InformMARL configurations across varying team sizes and sensing radii is computationally intensive, requiring GPU-accelerated hardware and parallelized rollout environments to complete within the REU timeline. Local training on a personal machine is not feasible at this scale. We request access to OSU's high-performance computing cluster for this project, with the expectation that training runs can be parallelized across configurations to minimize wall-clock time. *We look forward to discussing this proposal and welcome any feedback on the experimental design prior to implementation*

Works Cited

[1] S. Nayak, K. Choi, W. Ding, S. Dolan, K. Gopalakrishnan and H. Balakrishnan, "Scalable Multi-Agent Reinforcement Learning through Intelligent Information Aggregation," *Proceedings of the 40th International Conference on Machine Learning,* 2023.

[2] Z. Liu, Y. Li, J. Wang, J. Tu, Y. Hong, F. Li, Y. Liu, T. Sugawara and Y. Tang, "Robust and Efficient Communication in Multi-Agent Reinforcement Learning," *arXiv preprint,* 2025.

[3] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen and Y. Wu, "The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games," *Advances in Neural Information Processing Systems (NeurIPS),* 2022.

[4] R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel and I. Mordatch, "Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments," *Advances in Neural Information Processing Systems (NeurIPS),* 2017.